



US HOUSE PRICE ANALYSIS

Project Report I

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Group 08

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Abstract

The real estate market is a cornerstone of the U.S. economy. Housing prices are particularly important as they reflect broader economic conditions and impact financial markets. Understanding these factors is essential for buyers, sellers, and policymakers to make informed decisions. This report examines the impact of various property characteristics, and the spatial distribution of houses sold in the United States during 2014 on house prices. Through exploratory data analysis techniques, it was found that most observations clustered around Seattle. Evidence indicates that the majority of houses sold were priced within the mid to high range. One of the notable aspects is that buyers showed less interest in houses with basements and those with more than two floors. This trend is supported by the nature of the geolocation “the coastal area”.

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Introduction

In today's world, real estate is a worthy investment in any part of the world. The U.S. real estate market, in particular, holds a renowned place due to the country's immense wealth, its effect on the world economy and military power. Identifying key patterns in housing prices is essential for various stakeholders, including investors, buyers, and real estate developers. By understanding the dynamics of housing prices, investors can identify profitable opportunities, and buyers can make informed purchasing decisions. Our primary focus is to uncover potential patterns in housing prices, a crucial sub-sector of real estate that shapes investment decisions and market trends through an Exploratory Data Analysis (EDA).

This study is based on housing market data from 2014, providing insights into historical trends and potential factors affecting property prices. As of December 2024, the U.S. real estate market recorded 4.24 million sales, a median sales price of \$404,400, and an inventory of 3.3 months. By analyzing these patterns, we aim to identify key drivers of housing prices, offering valuable insights.

Description of the Question

The data set has been analyzed to explore the following three primary questions:

1. What are the key factors that influence house prices within this data set?
2. How do house prices compare across low, mid, and high ranges, and what features contribute to achieving the best price within each category?

3. In what ways has the location affected the built-in properties of the houses? Are there any significant patterns that emerge from this relationship?

Description of the Dataset

The following dataset sourced from Kaggle pertains to **HOUSE PRICES** in the United States, collected during the year 2014. This dataset includes 18 distinct variables. The data collection occurred between May 2 and July 7, during which a total of 4,600 house prices were recorded.

Variable	Definition
Date	The date the house was sold.
Price	The price of the house.
Bedrooms	Number of bedrooms.
Bathrooms	Number of bathrooms (1 = full bathroom, 0.5 = half bathroom, 0.25 = quarter bathroom).
Sqft_living	Living area in square feet (sqft_above + sqft_basement).
Sqft_lot	Lot size of land in square feet.
Floors	Number of floors (1 = full floor, 0.5 = small upper area).
Waterfront	Indicator (0 = No, 1 = Yes) for waterfront property.
View	Quality of the view (0 = No special view, 1 = Slightly better, 2 = Moderate, 3 = Good view, 4 = Excellent view).
Condition	Condition of the house (1 to 5 scale, higher is better).
Sqft_above	Square footage of the house above ground.
Sqft_basement	Square footage of the basement.
Yr_built	Year the house was built.
Yr_renovated	Year the house was last renovated (0 if never renovated).
Street	Street address.
City	City location.
Statezip	State and ZIP code.
Country	Country (all entries seem to be 'USA').

Table 1: Description of Variables

Data Preprocessing

Variable Redefinition and Assumptions

- The dataset originally lacked definitions for its variables. Hence this aspect was carefully done under based on assumptions derived from the real estate market in the USA.
- For the purpose of this analysis, fully renovated houses were treated as being comparable to new houses.
- The price variable was considered to be an ordinal variable with no temporal dependency.

Variable Creation, Editing and General Preprocessing

- The “date” variable has been edited to retain only month-day.
- The “country” & “street” variables have been removed.
- A new variable “House_Age” has been created, calculated as (2014 - yr_built) for houses that have not been renovated, and (yr_renovated - yr_built) for renovated houses.
- An ordinal variable “sqft_basement_categorized” has been created, consisting of 11 categories (0: no basement, 1 – 10: correspond to increasing basement sizes in 500 sqft increments)
- Observations with 0 for the price variable has been removed from this data set, resulting in an approximate 1% loss of information from a total of 4600 observations.
- 2 Observations which do not consist of bedrooms or bathrooms were removed.

Main Results of the Analysis

Univariate Analysis

Summary Statistics: For each variable, a summary of the main statistics was gathered, analyzing the data **with** and **without considering the outliers in the price variable**.

	mean_x <dbl>	median_x <dbl>	mode_x <dbl>	Q25 <dbl>	Q75 <dbl>	min_x <dbl>	max_x <dbl>
price	565392.49416	465337.5	300000	325000	661872.50	7800	26590000
sqft_living	2141.42270	1970.0	1940	1460	2620.00	370	13540
sqft_above	1831.50714	1597.5	1140	1180	2310.00	370	9410
sqft_basement	309.91556	0.0	0	0	600.00	0	4820
sqft_lot	14947.89694	7679.5	5000	5000	10964.00	638	1074218
sqft_yard	13703.56990	6413.0	3150	3910	9595.25	-19	1072878
House_Age	28.75944	20.0	11	9	40.00	0	114

	mean_x <dbl>	median_x <dbl>	mode_x <dbl>	Q25 <dbl>	Q75 <dbl>	min_x <dbl>	max_x <dbl>
price	565392.49416	465337.5	300000	325000.00	661872.50	7800	26590000
sqft_living	2051.65110	1940.0	1940	1450.00	2550.00	370	4360
sqft_above	1761.34056	1570.0	1140	1180.00	2250.00	370	4000
sqft_basement	280.27254	0.0	0	0.00	580.00	0	1500
sqft_lot	7425.33391	7200.0	5000	4800.00	9543.25	638	19877
sqft_yard	6229.73035	5946.0	3150	3658.75	8150.50	-19	18090
House_Age	23.44646	17.0	11	9.00	34.00	0	86

Table 2:(a)Summary Statistics with Outliers (b)Summary Statistics without Outliers

Price

- Low-priced homes: Typically below \$150,000.
- Mid-priced homes: Approximately \$150,000 to \$300,000.
- High-priced homes: Approximately \$300,000 to \$500,000.
- Luxury homes: Above \$500,000.

The price analysis was conducted based on the above defined categories. Out of the observations, only 51 are below \$150,000. 50% of the houses sold are priced under \$465,000,

and 75% of the houses sold are priced under \$670,000. However, there are 1739 Obs. With prices over \$500,000, which accounts for nearly 50% of the train set.

Bedrooms, Bathrooms and Floors

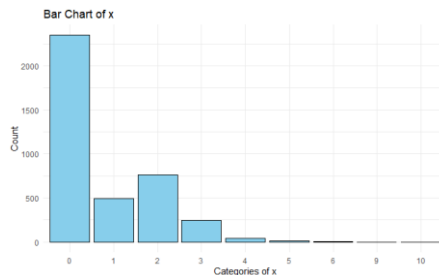
- Majority of the houses sold consist of 3 or 4 bedrooms, accounting for approximately 78% of the total observations. Even when excluding outliers, this percentage remains relatively unchanged at 73%. This suggests that the number of bedrooms does not significantly influence the other variables.
- 2.5 or less bathrooms are the most preferred, both with or without considering outliers.
- A fully setup floor (either 1 or 2 floors) was a feature of interest for 87% of the houses sold.

Residential Area Attributes

Almost 75% of the houses sold have **combined living area** (including both above ground and basement) of less than 2700 sqft as shown in Figure 1. Even though the upper 25% shows rapid distribution, it is primarily confined to less than 4000 sqft in size, even when accounting for outliers.

0%	25%	50%	75%	100%
370	1460	1970	2620	13540

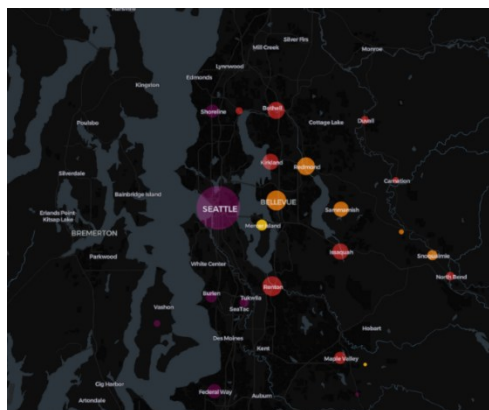
Figure 1: The Quantiles of sqft_living



Nearly 90% of the houses have an area of 2500 sqft or less above ground, and the sqft_basement indicates that most of the houses sold do not have a basement. This suggests that homeowners are not particularly concerned about having a basement for storage or any other purpose.

Figure 2: Bar Plot of sqft_basement_Categorized

Location Analysis



The map indicates that most of the data have come from specific areas, with Seattle identified as the major city from which the information has been gathered. Thus, observations cannot be generalized to all states in the United States.

Figure 3: Dense Cluster of Observations around Seattle (Resource: kepler.gl)

Bivariate Analysis

Association Between Price and Bedrooms, Bathrooms, and the Number of floors



Figure 4: Boxplot of Price vs Bedrooms

Boxplot of Price vs. Bedrooms: Mean house prices generally increase with the number of bedrooms, with the highest prices observed for houses with 6 to 8 bedrooms.

Boxplot of Price vs. Bathrooms: A positive correlation between the number of bathrooms and house prices can be identified, especially when the total exceeds 4 bathrooms.

Boxplot of Price vs. Floors: As the number of floors increases, there is a slight upward trend in house prices. Interestingly, houses with 2.5 floors (typically consisting of two full floors and a smaller upper area) exhibit the highest median price, surpassing those with both 2 and 3 floors. This suggests that the inclusion of a smaller upper area—often viewed as additional usable space or a premium feature may contribute to higher house values.

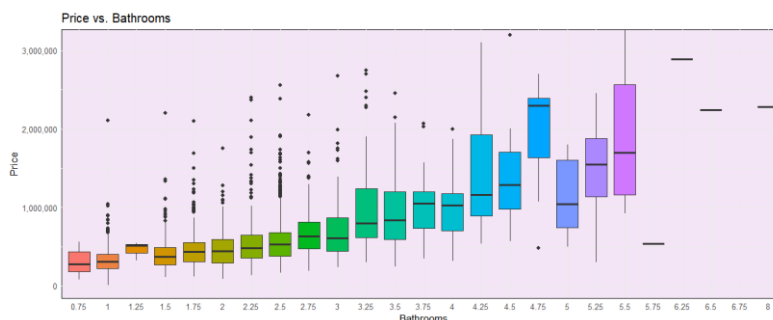
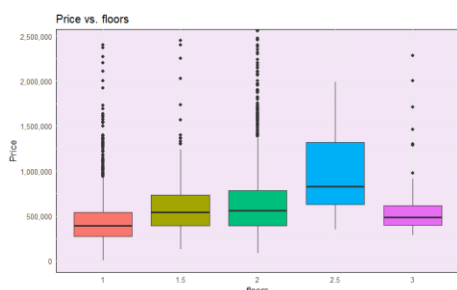


Figure 5: Boxplot of Price vs Bathrooms

Association Between Price and Waterfront, View, Condition, City

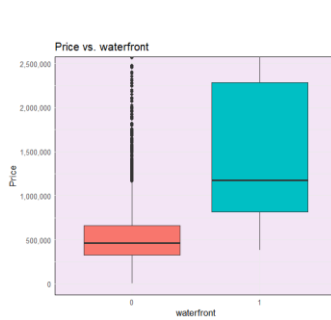


Figure 7: Boxplot of Price vs Waterfront

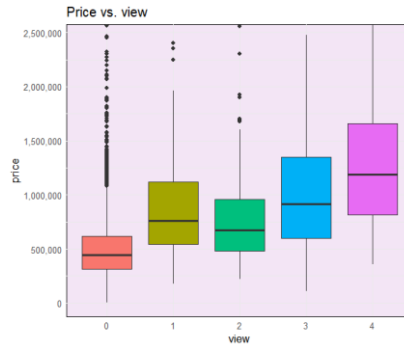


Figure 8: Boxplot of Price vs View

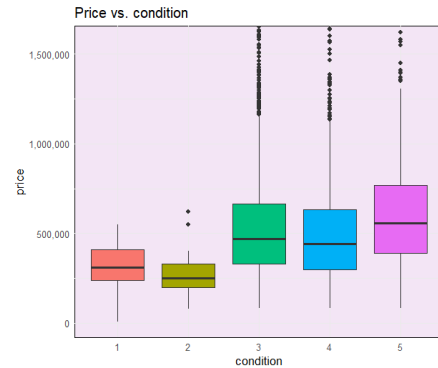


Figure 9: Boxplot of Price vs Condition

Boxplot of Price vs Waterfront: Houses located by the waterfront command significantly higher prices compared to those without. However, there are some outliers among non-waterfront houses with prices similar to waterfront properties. These exceptions may be influenced by other factors that affect house prices.

Boxplot of Price vs View: As the quality of the view increases, the mean house price also increases. A better view can significantly enhance the value of a house.

Boxplot of Price vs Condition: Houses in lower condition (levels 2 and 3) generally have lower mean prices, while those in higher condition (levels 4 and 5) tend to have higher prices. Additionally, houses with condition levels 1 and 2 show nearly the same prices, while levels 3, 4, and 5 are notably higher.

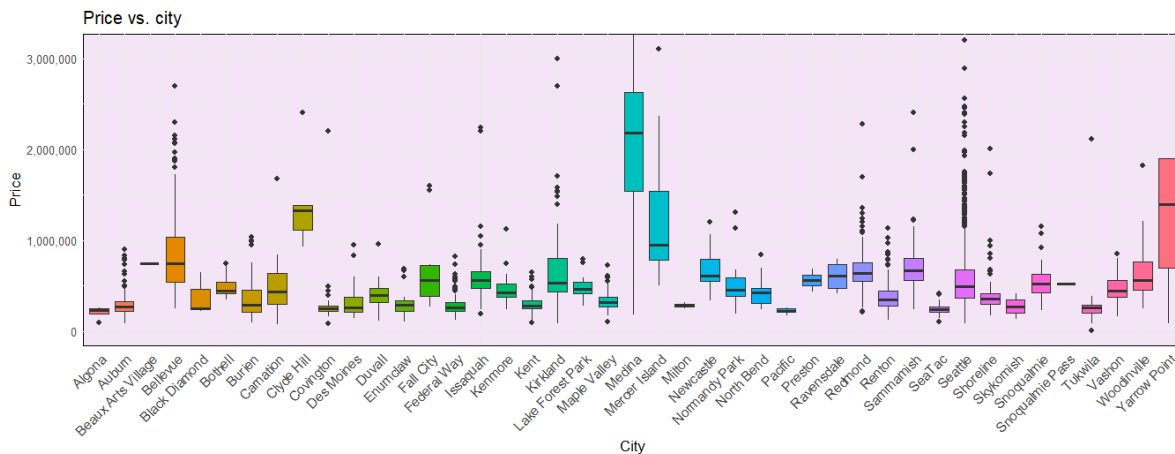


Figure 9: Boxplot of Price vs City

Boxplot of Price vs City: House prices vary significantly across different cities. Some cities have relatively low prices, while others are much higher. The city with the highest mean price is Medina, followed by Yarrow Point, Mercer Island, and Bellevue which also have high prices. In contrast, Milton and Altona have the lowest-priced houses.

Association Between Price and Square Footage Features

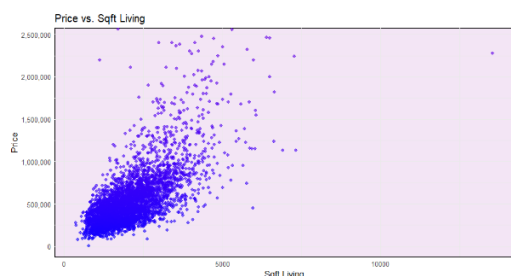


Figure 10: Scatterplot of Price vs Sqft Living

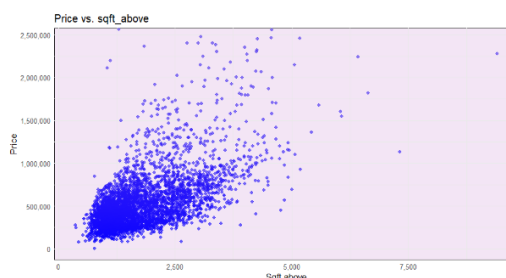


Figure 11: Scatterplot of Price vs Sqft Above

Price vs Square Footage: The above-ground square footage and living area are positively correlated with price, meaning that larger spaces tend to increase the price of the house.

Price vs Basement: As the size of the basement increases, the mean house price also rises. Houses with no basement (level 0) have an approximately similar mean price to those with small basements. However, houses with larger basements exhibit a significant price difference.

Multivariate Analysis

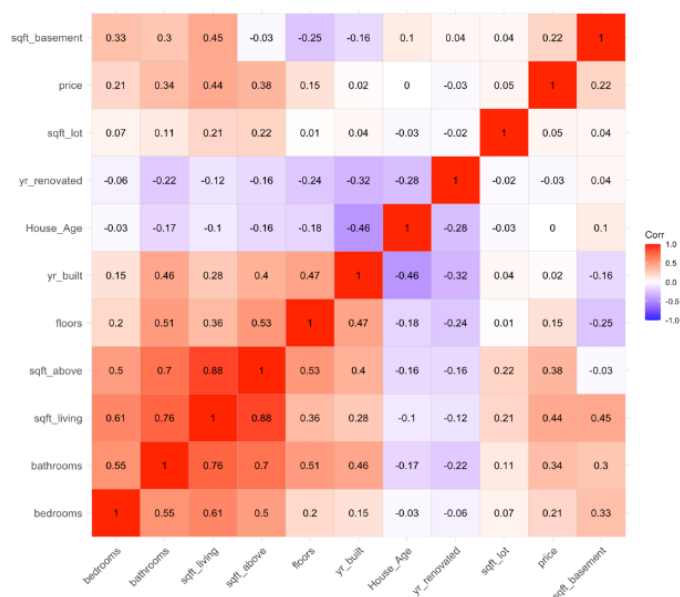


Figure 12: Correlation Heatmap

Among all variables, the total living area (sqft_living) stands out as the most significant variable, exhibiting the highest positive correlation with price. This signifies that buyers prioritize interior space over other elements such as the number of floors or the lot size. Moreover, features like the number of bedrooms, the number of bathrooms and the above-ground square footage show a moderate correlation with price, indicating that functional living space is highly valued by customers. Another fascinating finding is that there is no correlation between the age of the house (House_Age) and price. This reinforces the notion that factors such as renovations and the house's age have little impact on the market price.

Suggestions For Advanced Analysis

The Advanced Analysis is divided into 2 main parts.

- Supervised Learning
- Unsupervised Learning

Under supervised learning, the response variable will be the price, and the process will initiate with feature selection.

Multicollinearity and Outlier Handling

First, multicollinearity will be checked by using VIF, eigenvalues and condition number. Outliers will be addressed through transformations if necessary. Specifically, for skewed distributions, log transformation or Box-Cox transformations will be applied.

Testing of the Model

If multicollinearity is present, OLS methods may not provide significant insights. Therefore, Ridge Regression will be tested under the assumption that all variables remain in the model. If the Ridge model poorly performs on test set, Lasso Regression will then be tested followed by Elastic Net for further analysis. Residual Analysis will be conducted to uncover the underlying structures. If these models perform poorly, Principal Component Analysis, Factor Analysis and Multiple Correspondence Analysis will be performed to address evidence of multicollinearity. The next step will involve a thorough interpretation of the analysis results.

Cluster Analysis

The dataset naturally exhibits tendencies towards clustering due to the geospatial properties. Cluster analysis will be performed using **DBSCAN**. If the above mentioned methods perform poorly on the current test dataset, the methods will be revisited following the clustering analysis. Lastly, tree based methods and temporal dependencies by date will be tested as secondary performance measures.

Appendix

R Code

```

1 #DATA PREPROCESS
2 library(dplyr)
3 library(lubridate)
4 library(caTools)
5 #Import Dataset
6 library(readr)
7 data <- read_csv("C:/Users/ASUS/Downloads/data.csv")
8 #Change the directory
9 #View(data)
10 #Change data.csv into US_House_Price
11 US_House_Price <- data
12 rm(data)
13 #Count the no of zero price rows.
14 #Impossibility has been checked for zero values
15 sum(US_House_Price$price == 0)
16 sum(US_House_Price$bedrooms == 0)
17 sum(US_House_Price$bathrooms == 0)
18 sum(US_House_Price$sqft_living == 0)
19 sum(US_House_Price$sqft_lot == 0)
20 sum(US_House_Price$yr_built == 0)
21 ### Remove zero values
22 US_House_Price <- US_House_Price[US_House_Price$price != 0, ]
23 US_House_Price <- US_House_Price[US_House_Price$bedrooms != 0, ]
24 #checking for NA values
25 sum(is.na(US_House_Price))
26 # Create House_Age
27 US_House_Price <- US_House_Price %>%
28   mutate(House_Age = ifelse(yr_renovated == 0,
29     2014 - yr_built,
30     2014 - yr_renovated))
31 # Convert to Date-Time format
32 dates <- as.POSIXct(US_House_Price$date, format="%m/%d/%Y %H:%M")
33 # Extract Month and Date together (MM-DD format)
34 US_House_Price$date <- format(dates, "%m-%d")
35 # Removing 'country' and 'street' columns
36 US_House_Price <- US_House_Price %>% select(-country, -street)
37 # Checking for the number of duplicated values
38 duplicated_rows <- sum(duplicated(US_House_Price))
39 duplicated_rows
40 #sqft_basement_categorized
41 US_House_Price$sqft_basement_Categorized <-
42   round(US_House_Price$sqft_basement/ 100)*100
43 # Define bins and labels
44 breaks <- c(0,seq(1, 5000, by = 500),5001)
45 labels <- 0:(length(breaks) - 2)
46 # Categorize data
47 US_House_Price$sqft_basement_Categorized <-
48   cut(US_House_Price$sqft_basement_Categorized,
49     breaks = breaks, labels = labels, include.lowest = TRUE, right = FALSE)
50
51 #Dividing into train and test
52 set.seed(1234) #DO NOT TOUCH THE SEED UNDER ANY PURPOSE
53 sample <- sample.split(US_House_Price$price, SplitRatio = 0.8)
54 train <- subset(US_House_Price, sample == TRUE)
55 test <- subset(US_House_Price, sample == FALSE)
56 write.csv(train, "train.csv", row.names = FALSE)
57 write.csv(test, "test.csv", row.names = FALSE)
58
59 rm(US_House_Price);rm(US_House_Price);
60
61 attach(train)
62
63 library(ggplot2)
64 library(ggthemes)
65 library(scales)
66 library(ggcorrplot)
67
68 #UNIVARIATE ANALYSIS
69 #price-----
70 ggplot(train, aes(x = price)) +
71   geom_histogram(binwidth = 1000000, fill = "white", color = "black") +
72   labs(title = "Histogram of Prices", x = "Price", y = "Frequency") +
73   theme_gray()
74
75 Bp_1 <- boxplot(trainsprice)
76 Bp_1stats
77
78 Outliers <- Bp_1$out
79
80 sum(trainsprice>1500000)
81 sum(trainsprice < 500000)
82
83 #sqft_basement_categorized
84 # Convert to factor (ordinal)
85 sqft_basement_Categorized_N <- factor(sqft_basement_Categorized, ordered = TRUE)
86
87 # Create bar chart
88 ggplot(train, aes(x = sqft_basement_Categorized_N)) +
89   geom_bar(fill = "teal", color = "black") +
90   labs(x = "Categories of Sqft_basement_categorized",
91     y = "Count", title = "Bar Chart of Sqft_basement_categorized") +
92   theme_minimal()
93
94 #sqft_above-----
95 ggplot(train, aes(x = sqft_above)) +
96   geom_histogram(binwidth = 500, fill = "blue", color = "black") +
97   labs(title = "Histogram of sqft_above", x = "sqft", y = "Count") +
98   theme_minimal()
99
100 quantile(trainsqft_above, seq(0,1,0.1))
101 Bp_sqft_above <- boxplot(sqft_above$out)
102 quantile(Bp_sqft_above, seq(0,1,100))
103
104 # sqft_lot-----
105 hist(sqft_lot)
106 BP_2 <- boxplot(sqft_lot)
107 quantile(sqft_lot)
108 prob <- seq(0.75,1,0.01)
109 Qlot <- quantile(sqft_lot, probs = prob)
110
111 plot(prob,Qlot,type = "b",pch = 19,xlab = "Probability", ylab = "Quantile Value",
112   main = "Quantile vs. Probability Plot of 75%-100% percentiles")
113
114 #outlier analysis
115 BP_OUT_1 <- BP_2$out
116 hist(BP_OUT_1)
117 prob_out <- seq(0.1,0.1)
118 Out <- quantile(BP_OUT_1,prob_out)
119 print("Out");Out
120
121 #BIVARIATE ANALYSIS
122 # price with properties
123
127 ggplot(train, aes(x = factor(date), y = price, fill = factor(date))) +
128   geom_boxplot() +
129   theme_minimal() +
130   labs(title = "Price vs. date", x = "date", y = "Price") +
131   theme(legend.position = "none") +
132   scale_y_continuous(labels = comma) +
133   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
134
135 # Bedrooms
136 ggplot(train, aes(x = as.factor(bedrooms), y = price, fill = as.factor(bedrooms))) +
137   geom_boxplot() +
138   theme_minimal() +
139   labs(title = "Boxplot of Price vs. Bedrooms",
140     x = "Number of Bedrooms",
141     y = "Price") +
142   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
143   scale_y_continuous(labels = comma) +
144   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
145 # Focus on the 90th percentile
146 # Bathrooms
147
148 ggplot(train, aes(x = factor(bathrooms), y = price, fill = factor(bathrooms))) +
149   geom_boxplot() +
150   theme_minimal() +
151   labs(title = "Price vs. Bathrooms", x = "Bathrooms", y = "Price") +
152   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
153   scale_y_continuous(labels = comma) +
154   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
155
156 # sqft_living
157 ggplot(train, aes(x = sqft_living, y = price, color = price)) + # Color based on price
158   geom_point(alpha = 0.6) + # Add transparency for better visibility
159   theme_minimal() +
160   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
161   labs(title = "Price vs. Sqft Living", x = "Sqft Living", y = "Price") +
162   scale_y_continuous(labels = comma) +
163   scale_color_gradient(low = "blue", high = "red") +
164   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
165
166 # sqft_lot
167 ggplot(train, aes(x = sqft_lot, y = price, color = price)) +
168   geom_point(alpha = 0.6) +
169   theme_minimal() +
170   labs(title = "Price vs. Sqft Lot", x = "Sqft Lot", y = "Price") +
171   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
172   scale_x_continuous(labels = comma) + # Ensures real values in x-axis
173   scale_y_continuous(labels = comma) +
174   scale_color_gradient(low = "blue", high = "red") +
175   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
176
177 # sqft_above
178 ggplot(train, aes(x = sqft_above, y = price, color = price)) +
179   geom_point(alpha = 0.6) +
180   theme_minimal() +
181   labs(title = "Price vs. sqft_above", x = "Sqft above", y = "Price") +
182   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
183   scale_x_continuous(labels = comma) + # Ensures real values in x-axis
184   scale_y_continuous(labels = comma) +
185   scale_color_gradient(low = "blue", high = "red") +
186   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
187
188 # sqft_basement_categorized
189 ggplot(train, aes(x = factor(sqft_basement_Categorized), y = price,
190   fill = factor(sqft_basement_Categorized))) +
191   geom_boxplot() +
192   theme_minimal() +
193   labs(title = "Price vs. sqft_basement_Categorized",
194     x = "sqft_basement_Categorized", y = "Price") +
195   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
196   scale_y_continuous(labels = comma) + # Ensures real values in x-axis
197   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
198
199 # floors
200 ggplot(train, aes(x = factor(floors), y = price, fill = factor(floors))) +
201   geom_boxplot() +
202   theme_minimal() +
203   labs(title = "Price vs. floors", x = "floors", y = "Price") +
204   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
205   scale_y_continuous(labels = comma) +
206   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
207
208 # water front
209 ggplot(train, aes(x = factor(waterfront), y = price, fill = factor(waterfront))) +
210   geom_boxplot() +
211   theme_minimal() +
212   labs(title = "Price vs. waterfront", x = "waterfront", y = "Price") +
213   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
214   scale_y_continuous(labels = comma) +
215   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
216
217 # view
218 ggplot(train, aes(x = factor(view), y = price, fill = factor(view))) +
219   geom_boxplot() +
220   theme_minimal() +
221   labs(title = "Price vs. view", x = "view", y = "Price") +
222   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
223   scale_y_continuous(labels = comma) +
224   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
225
226 # condition
227 ggplot(train, aes(x = factor(condition), y = price, fill = factor(condition))) +
228   geom_boxplot() +
229   theme_minimal() +
230   labs(title = "Price vs. condition", x = "condition", y = "Price") +
231   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
232   scale_y_continuous(labels = comma) +
233   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
234
235 # city
236 ggplot(train, aes(x = factor(city), y = price, fill = factor(city))) +
237   geom_boxplot() +
238   theme_minimal() +
239   labs(title = "Price vs. city", x = "City", y = "Price") +
240   theme(
241     legend.position = "none",
242     panel.background = element_rect(fill = "#F3E5F5",
243       axis.text.x = element_text(angle = 45, hjust = 1, size = 10))
244   ) +
245   scale_y_continuous(labels = comma) +
246   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
247
248 # House Age
249 ggplot(train, aes(x = House_Age, y = price, color = price)) +
250   theme_minimal() +
251   labs(title = "Price vs. House Age", x = "House Age", y = "Price") +
252   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
253   scale_x_continuous(labels = comma) + # Ensures real values in x-axis
254   scale_color_gradient(low = "blue", high = "red") +
255   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
256
257 # yr_renovated
258 train$Renovated <- ifelse(trainsyr_renovated == 0, 0, 1)
259 upper_limit <- quantile(trainsprice, 0.95)
260 boxplot(price ~ Renovated, data = train,
261   names = c("Not Renovated", "Renovated"),
262   col = c("#E8F5E9", "#C8E6C9"),
263   main = "Boxplot of Price by Renovation Status (Zoomed In)",
264   ylab = "Price", xlab = "Renovation Status",
265   ylim = c(0, upper_limit))
266
267 # yr_built
268 ggplot(train, aes(x = yr_built, y = price, color = price)) +
269   geom_point(alpha = 0.6) +
270   theme_minimal() +
271   labs(title = "Price vs. yr_built", x = "yr_built", y = "Price") +
272   theme(legend.position = "none", panel.background = element_rect(fill="#F3E5F5")) +
273   scale_x_continuous(labels = comma) + # Ensures real values in x-axis
274   scale_y_continuous(labels = comma) +
275   scale_color_gradient(low = "blue", high = "red") +
276   coord_cartesian(ylim = c(0, quantile(trainsprice, 0.995)))
277
278 #Multivariate Analysis
279 # Correlation Analysis
280 # a correlation heatmap
281 num_vars <- apply(train, is.numeric)
282 train_numeric <- train[, num_vars]
283 train_numeric <- train_numeric[, setdiff(colnames(train_numeric),
284   c("sqft_basement_Categorized", "waterfront", "view", "condition"))]
285 cor_matrix <- cor(train_numeric)
286 ggcorrplot(cor_matrix, lab = TRUE, hc.order = TRUE)

```